Ex No: 14 | RAW sockets to packet sniffing

8.10.26

Aim:

To write a code using RAW sockets to implement packet sniffing.

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP,
                              UDP, ICMP
def packet-callback(packet):
    if IP in packet:
        ip-layer = packet[IP]
        protocol = ip-layer.proto
        src-ip = ip-layer.src
        dst-ip = ip-layer.dst
        protocol-name = ""
        if protocol == 1:
            protocol-name = "ICMP"
        elif protocol == 6:
            protocol-name = "TCP"
        elif protocol == 17:
            protocol-name = "UDP"
        else:
            protocol-name = "Unknown Protocol"
        print(f"Protocol: {protocol-name}")
        print(f"Source IP: {src-ip}")
        print(f"Destination IP: {dst-ip}")
        print("-"*50)
sniff(iface = "Wi-Fi", prn = packet-callback,
      filter = "ip", store = 0)
```

Output:

Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6
- - - - - - -
Protocol: UDP
Source IP: 192.168.1.6
Destination IP: 192.168.1.1.

Result:

The packet-sniffing program was successfully implement using scapy to capture network packets and display protocol types and IP addresses.